

Polymer Processing 1

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Aims of session & Learning outcomes



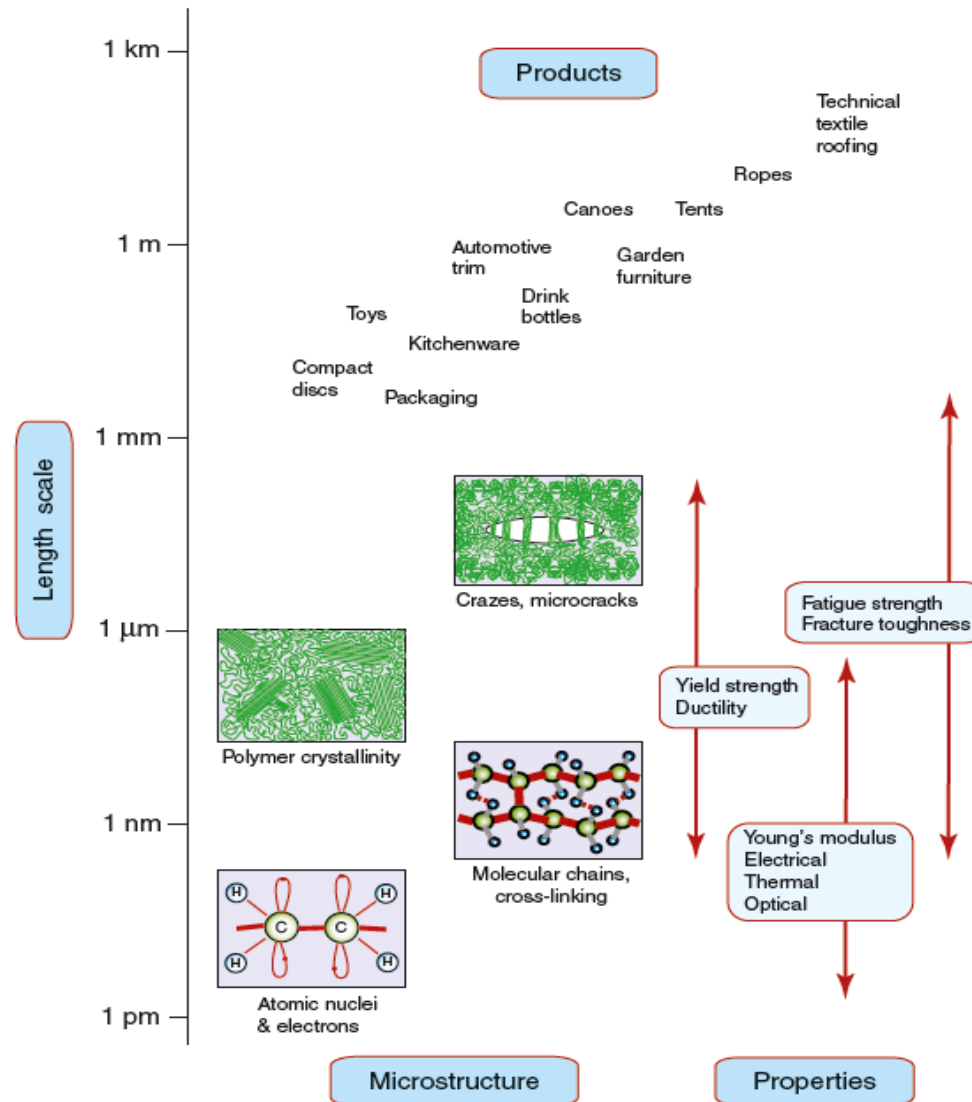
- Describe the role of additives in the fabrication of plastic
- Describe techniques for the fabrication of polymeric materials
- Explain the selection of fabrication technique based on the polymer type
- List, sketch and describe techniques for the fabrication of polymers

Polymer processing considerations

- From polymers to products - Additives
- Compounding of polymers
- Non-thermal processing
(thermosets and solvent processes)

Polymer processing 2 – thermal processing

Polymer and elastomers



- The role of **shaping processes** is to produce the right shape with the right final properties
- The first is achieved by controlling **viscous flow** or plasticity
- The second requires control of the nature and rate of **microstructural evolution**

Processing of Polymer and Elastomers

- Processing can determine the final molecular arrangement and properties of polymers
- Molecules can be aligned mechanically by the flow induced during the shaping process
- The most dramatic impact of processing on polymer properties is in making fibers
- Drawing fibers creates significant molecular alignment. The covalent bonding along the chain is exploited to markedly increase the specific strength and stiffness of the polymer

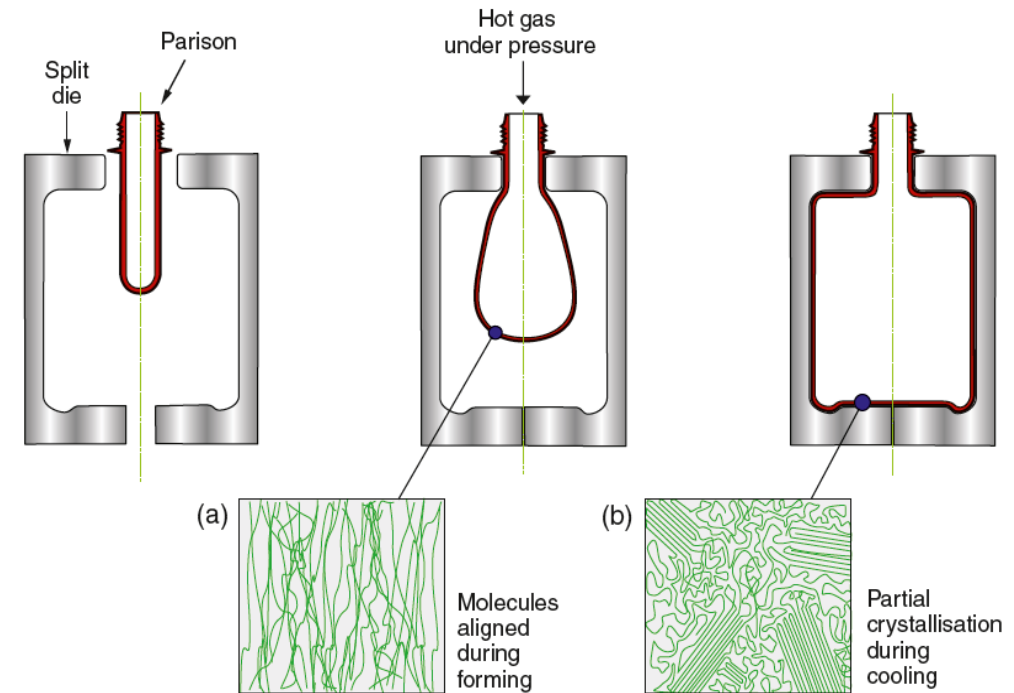


Figure 19.23 Evolution of molecular architecture in polymer moulding: (a) alignment of molecules during viscous flow in shaping; (b) partial crystallisation during cooling.

Processing of polymers

The method used for a specific polymer depends on several factors:

- 1) If the material is thermoplastic or thermosetting
- 2) if thermoplastic, the temperature at which it softens
- 3) the atmospheric stability of the material being formed
- 4) The geometry and size of the finished product.



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Characteristics of Plastics that determine processing techniques



Thermoplastic

- Can be reversibly cooled & reheated, i.e. recycled
- Heat until soft, shape as desired, then cool
- Examples: polyethylene, polypropylene, polystyrene.

Thermoset

- When heated forms a molecular network (chemical reaction)
- Degrades (doesn't melt) when heated
- A prepolymer molded into desired shape, then chemical reaction occurs
- Examples: urethane, epoxy

From polymers to products

Most engineering polymer products are **not just made of polymer!**

The polymers need other compounds to make useful formulations for application as durable engineering materials.

Additives:

- Fillers
- Plasticizers
- Colour
- Stabilizers
- Catalysts
- Foaming agents

Mixed (compounded) with the polymer before forming

Fillers

Added to polymers to:

- Increase stiffness
- Increase strength
- Increase wear resistance
- Reduce thermal instability
- Reduce cost

Filling materials may be:

- Wood flour (waste material)
- Fumed silica (waste material)
- Sand (natural silica)
- Calcium carbonate
- Glass
- Mica
- Carbon black

Filling materials are added as fine powders to the polymer

Larger particles ($> 10\mu\text{m}$) might be considered as particle composites

Carbon black (10-100nm) is the most important in engineering elastomers

Some fillers might also have other functions in the plastic e.g. colour, UV absorption

Plasticizers

Small molecules (low Mw polymers and oligomers)

They are mixed with high Mw polymers ~ 30%wt to **improve flow** and **reduce brittleness**

Plasticizers must be **miscible** (mixable, have similar intermolecular forces) with the polymer. They must also have a low vapour pressure and diffusion rate in the polymer, so that they are **not lost** from the polymer

The molecules make space between the polymer chains – **reducing T_g**

They can make a glassy polymer (e.g. PVC $T_g = 87^\circ\text{C}$) soft and plastic at room temperature. They will also lower processing temperatures.

Non-polar polymers (e.g. polyethylene) are difficult to plasticize

Colour

Colour or pigments are often added to polymer materials so that the final product does not need any other finishing or painting

Colourings may be organic or inorganic:

- White = TiO_2 (also good UV reflector)
- Black = carbon black (also good UV absorber)
- Glitter = mica
- In-organic minerals (different colours)
- Organic dyes



The colouring may also act as a plasticizer or a filler and change the properties of the polymer, e.g. by interrupting crosslinking or initiating crystallization

Stabilizers to prevent degradation from:

- Oxidation – or hydrolysis of polymers with double bonds
- Photo-chemical degradation – UV light absorption
- Free radicals – to absorb left over initiators
- Fire – flame retardants

Catalysts

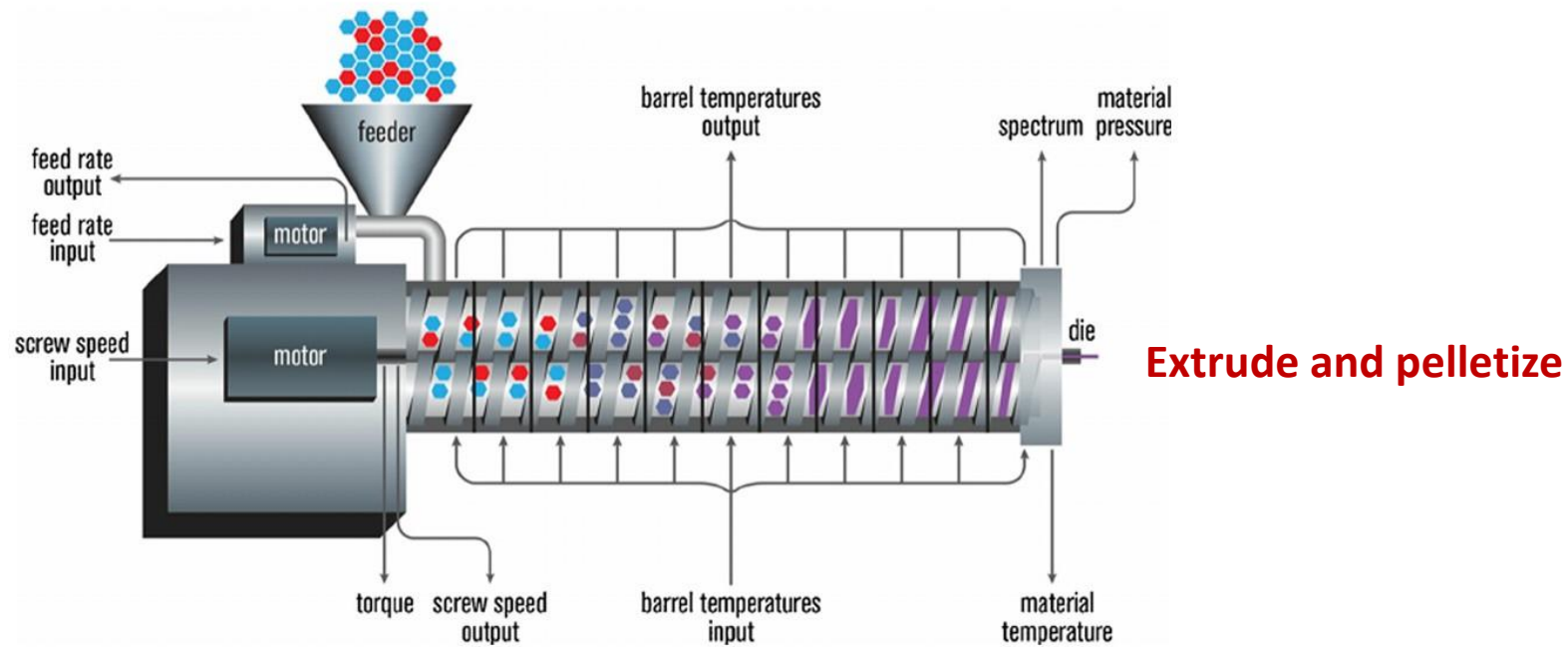
- To initiate cross-linking
- To initiate crystallization

Foaming agents

- To generate gas during processing to make a polymer foam

Compounding

Polymers and polymer blends may be mixed with additives before the forming process – using a **twin screw extruder**



Extrude and pelletize

Heating Mixing Homogenise

- The 2 screws turn at different rates and pitches to create high shear and mixing.
- The extruded polymer is chopped into pellets for further processing into the product

Compounding

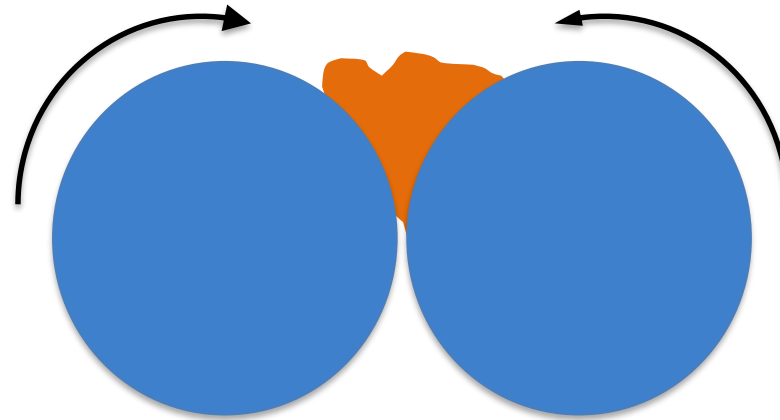
2-roll mill

Elastomers are often compounded in a 2 roll mill before vulcanization (cross-linking with sulphur) and shaping in a heated mould

The 2 rollers turn at slightly different speeds to create high shear and mixing

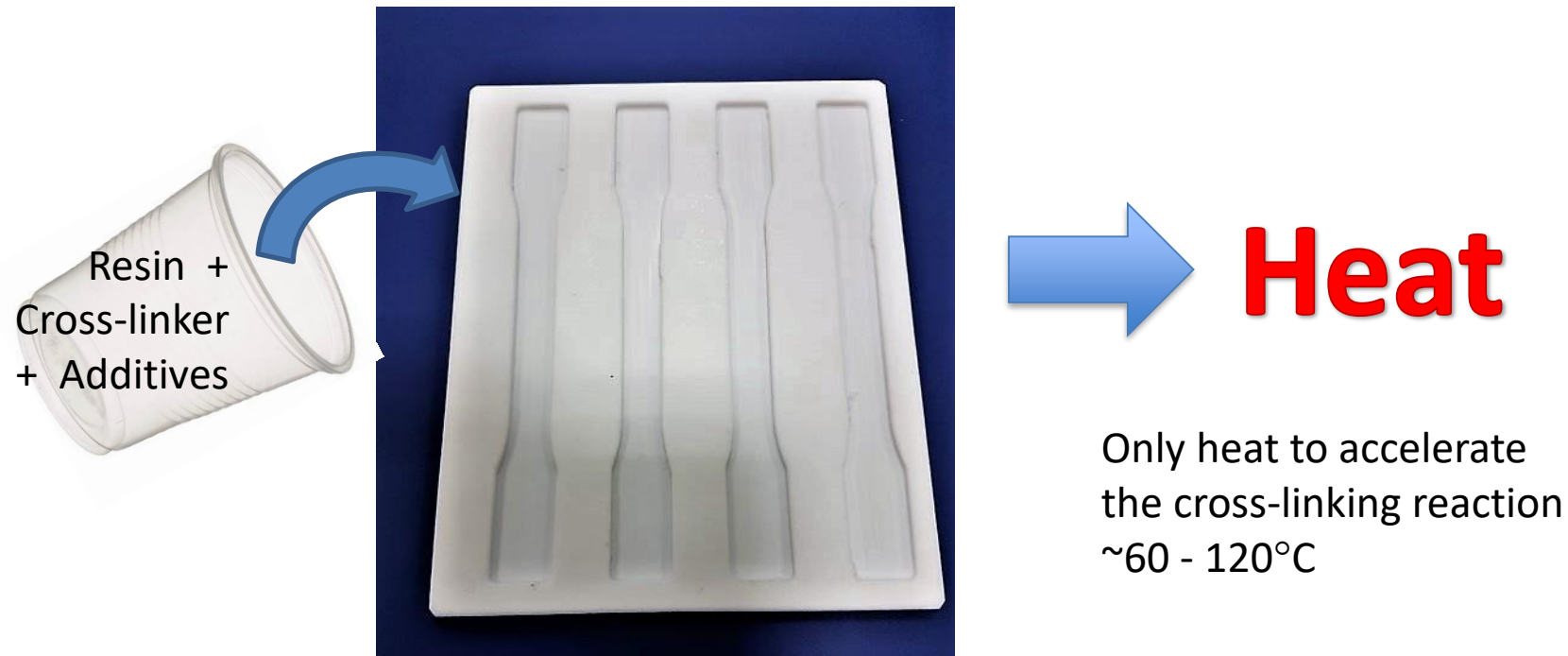
Rubber compounding

- Natural rubber
- Sulphur cross-linker
- Carbon black filler
- Catalyst (initiator)



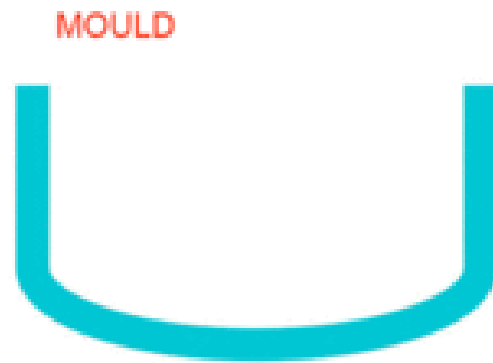
Thermosets and cross-linked elastomers

- Mix polymer and cross-linker + additives
- Pour into mould to form the required shape
- Heat mould to initiate cross-linking (thermally activated)



Rotational moulding or Roto-casting

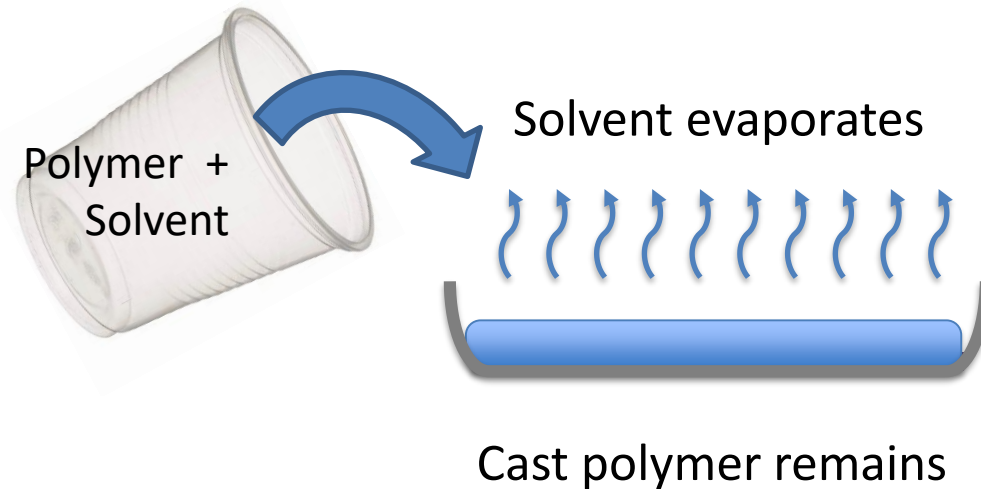
- Hybrid of casting and moulding
- Thermoplastics – and thermosets
- Melt thermoplastic to 'cast'
- Heat thermoset to cross-link



Example:
large tank or container

Solvent casting

- Polymer mixed with solvent to make low viscosity liquid
- Pour into mould and allow solvent to evaporate
- Some crosslink reactions may be enabled by the loss of solvent



Used to make:

- Coatings
- Thin polymer films
- Membranes

**Problem for clean up:
Polymer often left on surfaces!**

Engineering plastics are not just polymers!

Additives use to make processing easier and more durable product

Polymers are often compounded together before final forming

Compounding (mixing and formulation) to make raw material

For further processing into product:

- Casting
- Thermal processing